

Wifi Walkie

What is it?

Wifi Walkie is an audio communication app built in Kotlin. It allows two devices to connect over LAN or WAN and users can speak via push to talk. While walkie talkies are traditionally half-duplex, we will be designing for full-duplex with half-duplex as an optional setting.

What does it solve?

The app solves the need for fast, push-to-talk voice communication between nearby users that are connected through the same Wi-Fi network. It provides a walkie-talkie style communication and experience without requiring the traditional radio hardware or cellular service.

Classes and Functions

Connection manager

- User connection
- User disconnect
- Password/Passkey for connection

Audio stream

- Start
- Stop
- Pop

Audio sender

- Transmission beep
- Send audio through connection manager

Audio receiver

- Transmission beep
- Receive audio through connection manager
- Send received stream data to audio player

Audio player

- Play

Stretch goals

- iOS port
- Half-duplex mode
- Screen-off mode, where transmission is driven by the volume keys
- Peer-2-peer

- Vibration transmission for morse code
- Encode communications for security
- “Repeat that” feature